https://github.com/Et1enneee/comp3500-logistics

COMP 3500SEF · Final Report

Warehouse Logistics Management System
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1. Scope

The goals are traceable order decisions and a role-separated workflow. Customers order and track; staff confirm, pick, ship and assign; drivers simulate assigned deliveries; administrators report. The local application uses Next.js, TypeScript, Prisma and SQLite. GPS, scanning, telephony, payment and carrier integration are outside scope. Deployment is local-only; data integrity is enforced by transactional guards (§5).

2. Development process

Plan-driven increments have triggers and verified outcomes; integration and continuous integration provide feedback. Increments cover product and specification changes and recorded process milestones.

Table A. Increment, trigger and outcome

Increment and date	Trigger	Outcome
Planning, 9 and 11 Oct 2026	Project selection	Responsibilities and contracts recorded
17 Oct test-led review	Assess order and stock functions	Defects recorded in the register; led to the confirmation transaction (§5)
Confirmation, 26–28 Oct 2026	Failed endpoint and missing stock ledger (register)	Transaction verified (orders tests)
Integrated application, 1 Nov 2026	Integrate foundation	Customer, staff and admin workflows
Driver and privacy, 2 Nov 2026	Delivery and data minimisation	Assigned tasks and data-minimisation checks (tests)
Documentation hardening, 10–11 Nov 2026	Model and evidence review	Diagrams and traceability aligned (traceability)
Export simplification, 11 Nov 2026	11 Nov scripted walkthrough (issue #15)	Selected-date export in three actions; regression passed
Availability tests, 11 Nov 2026	Regression gap in availability checks	Suite increased to 38

3. Requirements

Functional requirements and verification

FR	The system shall	Verification
FR-01	create orders with product lines and totals	Test — creation and stock
FR-02	enforce lifecycle and stock reservation or release	Test — lifecycle and stock
FR-03	number orders and shipments and show owner tracking	Test — identifiers and tracking
FR-04	show inventory, replenishment, ledger (inventory movement log) and low-stock alerts	Test — inventory and alerts
FR-05	record event notifications and allow reading	Test — notifications and reads
FR-06	provide admin aggregates and CSV export	Test — aggregates and export
FR-07	authenticate and enforce role and object access	Test — authentication and boundaries
FR-08	assign shipped orders and simulate delivery	Test — assignment and delivery

NFR-01 performance (API p95 ≤2 s, first contentful paint ≤3 s); NFR-02 integration availability (green build–test–browser runs recorded per increment); NFR-04 ≤3 actions per scripted operation.

NFR-03 least privilege — role gates on every mutating endpoint, verified by auth tests; NFR-05 process quality — CI-verified increments (workflow).

Table B. Risk coverage (detailed mapping)

Risk	Addressed by	Evidence
RK-01 scheduling slip	Increment register	scheduling plan
RK-02 contract rework	FR-02, NFR-05	transition tests
RK-03 test-week capacity	Priority register	scope priorities
RK-04 setup delay	NFR-05	CI workflow
RK-05 demo data	FR-01/04/06, NFR-05	seed regression test
RK-06 assignments and test squeeze	Risk plan	assessment schedule
RK-07 stack unfamiliarity	NFR-05 support	setup guide
RK-08 report and video compete for time	Risk plan	report and video plan

4. System models

Local transactions preserve consistency without splitting services; state and stock guards enforce rules. The four layers — Role-specific pages, HTTP routes and authentication, Order/inventory/driver business handlers, Prisma/SQLite persistence — each isolate one concern.

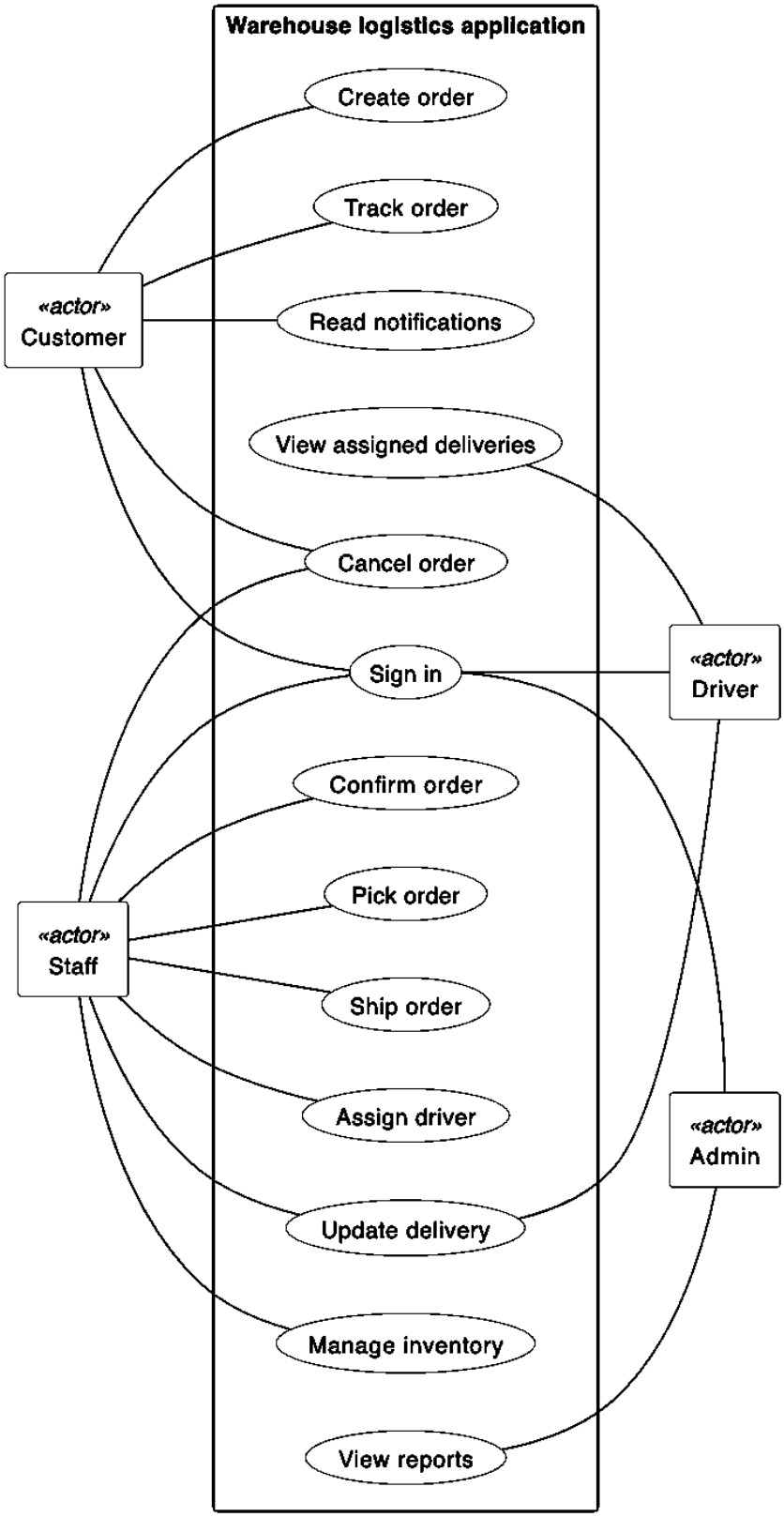


Figure 1. Which roles perform each function? Source.

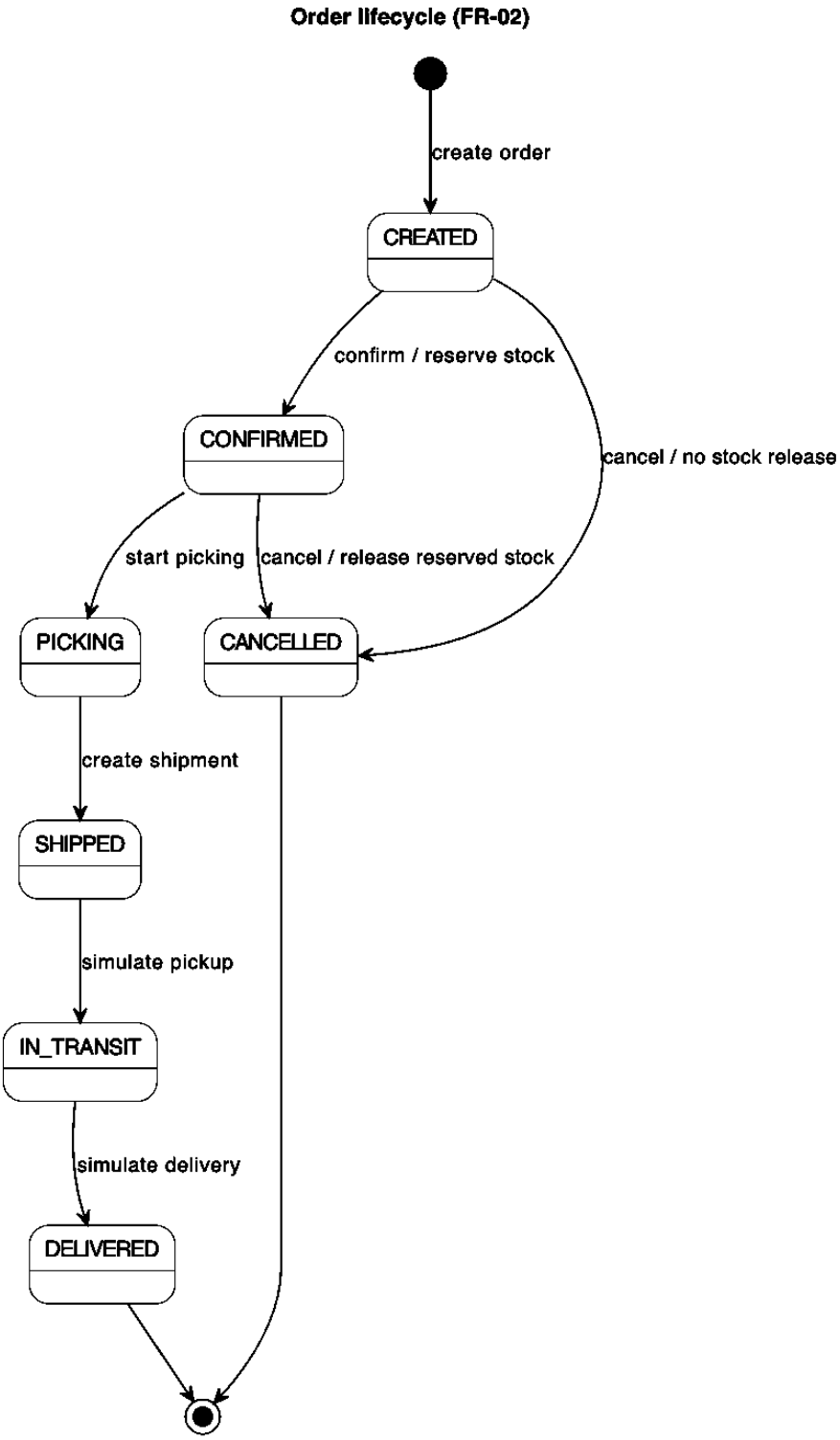


Figure 2. Which order transitions and stock effects are valid? Source.

Each item references one product; each order requires at least one item; each order has 0..1 shipment; each product has 0..* inventory logs.

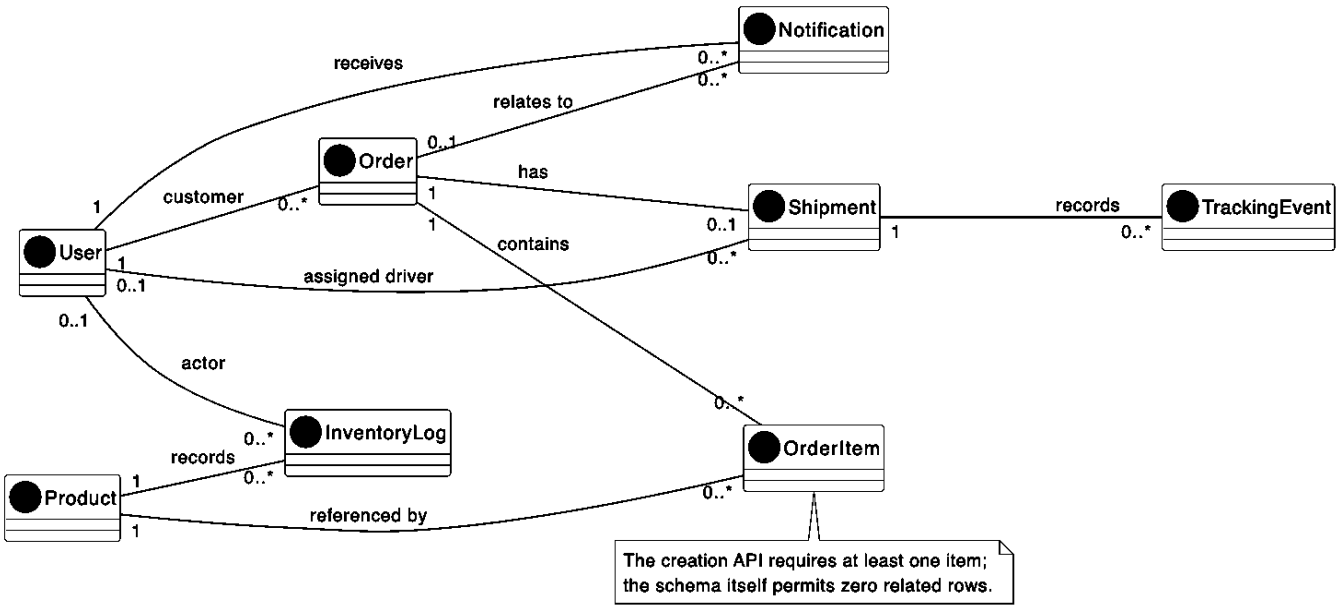


Figure 3. Data relationships: each order requires at least one item, enforced at the creation API. Source.

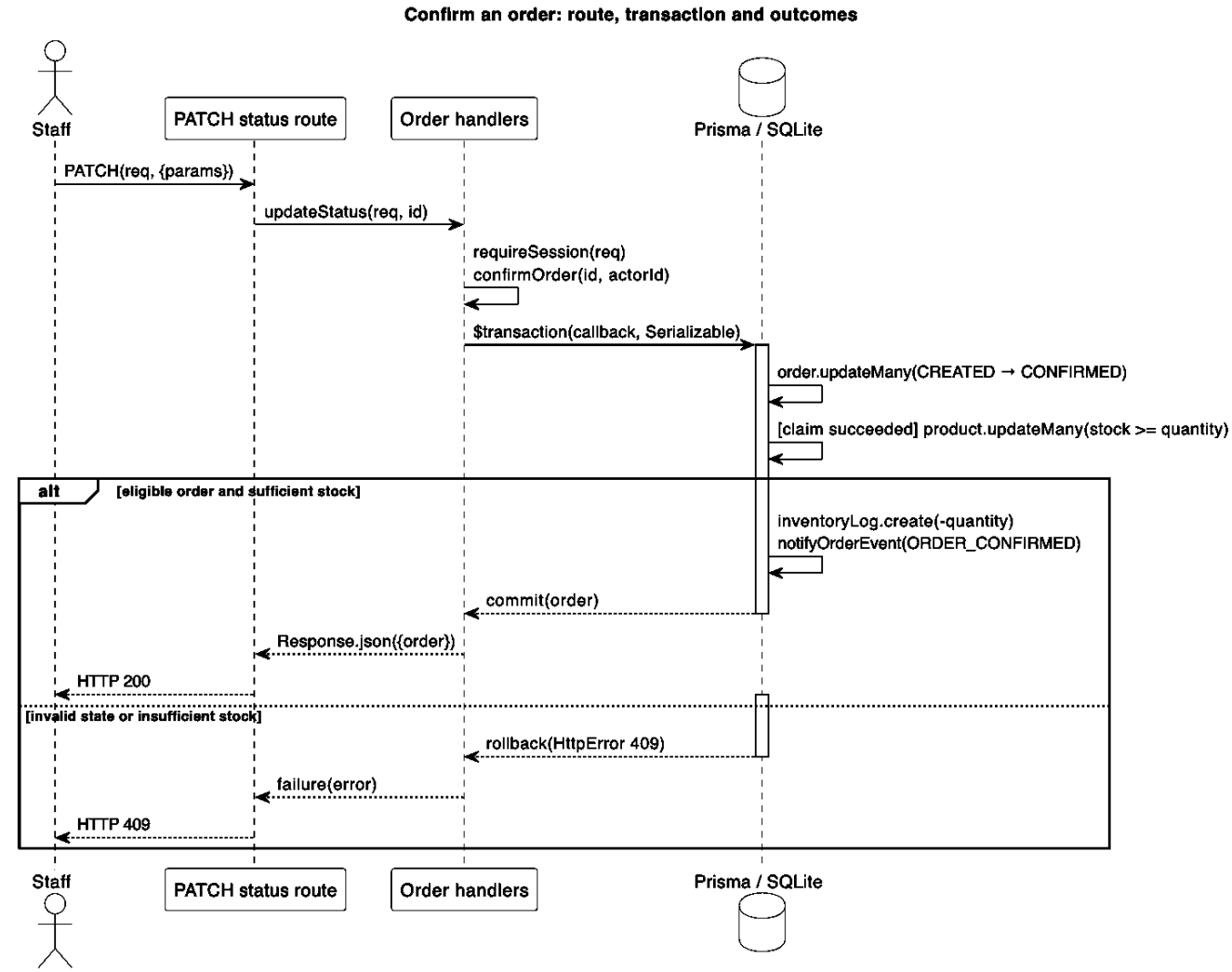


Figure 4. How does confirmation commit or roll back? Source.

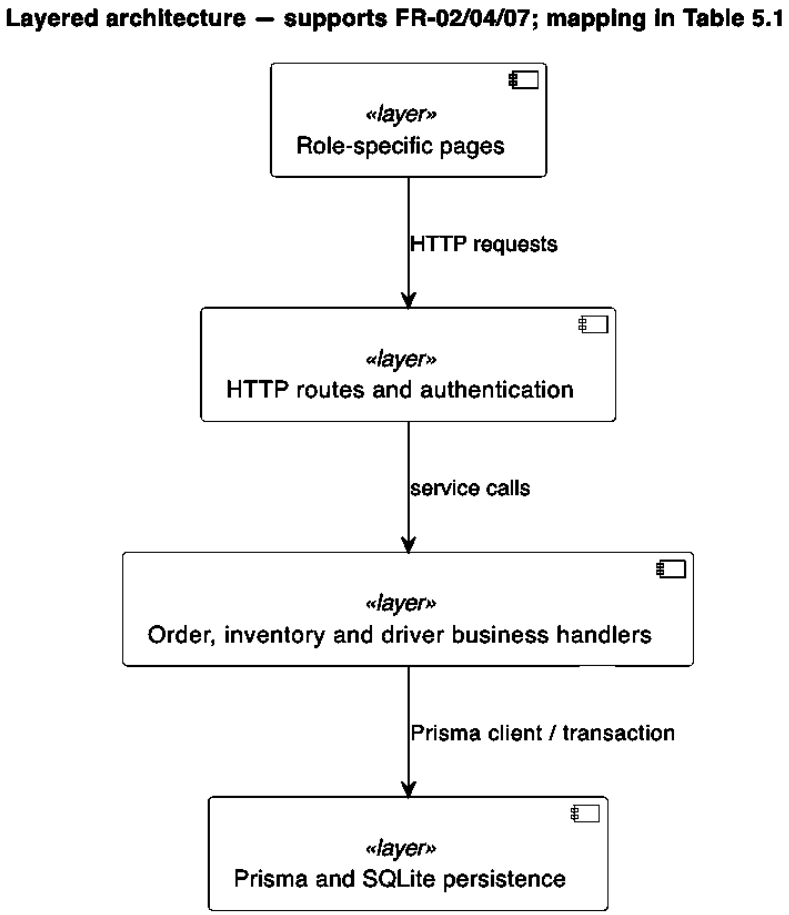


Figure 5. How do four layers connect? Source.

5. Implementation

Cancellation:

Role	State	Effect
Customer	CREATED	cancel, no restock
Customer	CONFIRMED	cancel, restock
Staff	CONFIRMED	cancel, restock
Both	PICKING onwards	rejected

Design	Implementation	Verification
Guards: CREATED and sufficient stock	order-api.ts:250–253, 276–280	Orders
Serializable state, stock, ledger and notification writes	order-api.ts:245–293	Orders
driverId: assign only SHIPPED; retained after delivery	driver-api.ts:44–51; transition	Driver

6. Verification and validation

Verification checks requirements against the implementation; validation exercises the delivered workflows through the interface. The regression log and CI show 38 unit, 140 integration, 2 delivery checks (assignment and delivery simulation) and 26 browser checks.

Insufficient stock returns HTTP 409 with unchanged state. In a local-loopback measurement recorded on 11 November 2026, API p95 was 3.2–13.6 ms and first-contentful-paint p95 was 64–96 ms, against NFR-01 targets of at most 2 s and 3 s respectively. Targets are end-to-end budgets; loopback bounds server-side processing. NFR-02 is evidenced by the recorded green build–test–browser runs (observations).

7. Evolution

The 17 Oct test-led review found a failed confirmation endpoint and a missing stock ledger; the confirmation transaction and its retest followed (26–28 Oct). The 10–11 Nov documentation review corrected the diagrams and aligned traceability against the reviewed draft. Two availability regression tests added on 11 Nov brought the suite to 38.

8. Validation

The automated walkthrough exercises eleven core operations through the interface as each of the four roles, at most three actions each, with expected outcomes asserted in the walkthrough script; results are logged with the walkthrough record; the eleven operations map one-to-one to the 26 browser checks, expected outcomes asserted in the script.

9. Feedback and reconciliation

Table C. Feedback to evidence

Feedback	Disposition	Source	Eviden
Diagrams	Corrected models and architecture	10–11 Nov documentation review	Compli
FR-04	Stock transition effects belong to FR-02	17 Oct test-led review	Traceal
Increments	Triggers and outcomes	10–11 Nov documentation review	Registe
Admin task	Export re-tested; walkthrough records the task (§8)	11 Nov scripted walkthrough	Issue #
Unit count	Two availability regression tests added, bringing the suite to 38	regression runs	Test his
NFR scope	Performance and availability criteria set for local single-warehouse deployment	regression runs	SRS re

10. Evidence

Traceability · Baseline record.

Evidence index; baseline `6eb01afb1798c7361205b10597d80c0c26e8fdf6`.